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The Ethics of Privacy and Data Handling in Computer Science

Computer Science is faced with many different ethical issues. One of the most major of these is the way in which data is to be handled. A lot of work in the field of Computer Science involves the use of data from a variety of different sources, and often this data is considered private and confidential. Privacy is, or at the very least should be, extremely important to anybody who uses computer systems that use their data. Whether it is online banking, shopping, or even social media, a person has the right to keep their information private. As a future professional within Computer Science, I believe that the most significant ethical issue I will face in my field is pertaining to data security and privacy and properly managing and protecting the data of any user of the system.

Modern computer software is often collecting extreme amounts of data, and in most cases this data extends far beyond the users' knowledge of it. Researchers from the IEEE quote that when thinking about data privacy, individuals should “Do no harm while seeking to improve the quality of life for all people” (Balance Ethical and Legal Requirements). As a professional, I will likely be tasked with overseeing a system that utilizes user data without their knowledge. This can raise some ethical concerns about autonomy and consent to data. While data driven design is often the norm for computer systems, if it is misused then it could be harmful to the user. As

stated by the IEEE, keeping the users' best interests in mind should be the first step to approaching the ethical issue of data privacy.

Another ethical challenge when thinking about data privacy is how secure the data is within the system. Information collected from the user should only be stored while it is still required. The ACM code of ethics says that, in refence to data collection, “retention and disposal periods for that information should be clearly defined, enforced, and communicated to data subjects” (ACM). Unchecked data collection can become harmful to the user, as their personal and sensitive information risks exposure to data breaches, or could even be exposed to the developers within the system. If my work in the future involves managing data systems, it will also be my responsibility to make sure that data disposal is managed and completed successfully. The IEEE states that we need to work to balance the legal requirements of data handling and the morality of the issue (Building Trust in the Information Age). As a computer scientist, it will ultimately be my responsibility to be aware of these issues and properly work through them with all these ethical dilemmas in mind. I will likely face situations where these ethical dilemmas are present, and I must handle them properly, so the privacy and safety of others are properly maintained.

To face these ethical challenges, a person must be able to identify how the user could potentially be harmed by certain systems, and they must be able to keep these issues at the forefront of their mind when managing data. I believe that a mindset that puts the people, or the user, first and foremost, will be the most beneficial to the people navigating this issue. I am not fully prepared to address these challenges, as I still have many things to learn before I am knowledgeable enough to understand fully how to tackle all of these issues within the workplace.

I personally believe that to prepare myself, I will need to become more knowledgeable in the field of cybersecurity, so I will be able to properly protect the user.

The ACM code of ethics provides a stable view on professional work ethics and other principles. These principles also happen to align well with biblical principles. For example, principle 1.6 of the ACM code talks about the respect of privacy, and it explains how the privacy of other people deserves to be respected. This principle reflects biblical values, as the Bible also teaches that we followers of Christ should be trustworthy people. The book of Luke says, "The good person out of the good treasure of his heart produces good, and the evil person out of his evil treasure produces evil, for out of the abundance of the heart his mouth speaks." This verse teaches us that whatever is in our heart will also show in our works, including our duties as employees. Therefore, it is important to be of good character, so that the goodness of our heart will show in our work. The book of Matthew says, "So whatever you wish that others would do to you, do also to them." Nobody wants their data to be misused, so it is important to manage other people's data properly as well. The ACM also states that information should only be taken without violating the individual's rights (ACM). Both the ACM code of ethics and the Bible talk greatly on these topics of privacy and respect, and it is clear to see just how much they both support each other.

The ethics of data privacy and security is one of the most important ethical issues I will face in the Computer Science field. While data collection is necessary in most fields of Computer Science, it can very easily be handled unethically. Dilemmas from data privacy include keeping users informed of how their data is used and protecting their data from potential data breaches and cyberattacks. It is important to keep ethical codes like the IEEE and the ACM code of ethics, and the principles taught in the Bible in mind so that these situations are handled properly and

ethically. By keeping these things in mind, myself and professionals within the field of Computer Science will be able to properly navigate these ethical issues and keep the best interests of the user in mind.

Works Cited

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